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The Editor-in-Chief

Pattern Recognition Letters

Re: PRLETTERS-D-26-02906, revised submission

Dear Editor,

Thank you for the editorial office's message of 9 November 2026. In response, I have uploaded the Authorship Confirmation form as a Cover Letter item, and I have reformatted the manuscript to the journal's requirements: elsarticle layout, a length well inside the seven-page limit including figures, tables and references, an abstract of fewer than 200 words, five highlights of at most 85 characters, author-year references and high-resolution figure files.

I have also changed the manuscript itself, and the title with it. The first version, "MyUncleVision: An Interpretable Lattice-Based Framework for Unsupervised Anomaly Discovery in Whole-Slide Histopathology Images", was written around one application and was far longer than the limit. The revised manuscript, "Order and entropy of lattices of unit vectors in d dimensions: label-free discovery of unusual local organization in simulated lattices and synthetic images", puts the method first and condenses the evidence into a letter.

The manuscript extends the local order and entropy quantities of the MyUncle maintenance-lattice model from the plane to unit vectors in d dimensions, using the von Mises-Fisher distribution; they equal MyUncle's at d = 2, which I tested against the unmodified code. It builds a label-free score that judges how rare the state of a site is at several prevalence scales and smooths the score over the lattice. The score is compared with isolation forest, local outlier factor, nearest-neighbor and Mahalanobis baselines, with settings frozen before the held-out data were generated, on two testbeds: lattices simulated with MyUncle (damaged sites as ground truth) and synthetic stained-tissue images.

The study states plainly what it shows and does not show. The results are mixed: lattice smoothing lowered the average precision of every baseline on the simulated lattices because the damaged sites are not compact regions, whereas on the synthetic images it raised the spatial coherence of every scorer; the frozen score was the best method with DINOv2 embeddings and a poor one with handcrafted features, and a simple Mahalanobis baseline was a strong competitor. All the evidence is simulated and was written by me, and no experimental data were used; the paper says so and lists validation on experimental micrographs and on histology as planned work. I submit it as a short methods contribution, and I would welcome the editor's view on whether that scope is suitable for the journal.

The revision also tests a supervised configuration search with GaaR, an evolutionary search framework developed by the author, on fresh test scenarios that no search saw, with paired 95% intervals obtained by resampling independent images and simulation runs. The search raised the lower-tail average precision from 0.18 to 0.26, but the interval of the difference includes zero and a plain random search also beat the default configuration, so any gain belongs to the configuration that was selected and not to the search algorithm. A controlled study on an independent embedding-lattice generator, which varies only the size of the anomalies, found that lattice smoothing hurt at every size and least for compact anomalies. The manuscript states that the selection is supervised, not label-free, and that all evidence is simulated.

Earlier versions of this work are openly archived on Zenodo (https://doi.org/10.5281/zenodo.24066256, all versions) and the code is open at https://github.com/sandlerleon/MyUncleVision (archived at https://doi.org/10.5281/zenodo.24066260). A ChemRxiv posting of the present manuscript is planned and will state that the paper is under consideration at Pattern Recognition Letters; I will tell the office if it is posted. The manuscript is not published and is not under consideration elsewhere, and it is not related to any conference paper.

A generative AI tool (Claude, Anthropic) assisted with writing and testing the code, running the analyses and drafting the text and figures; I reviewed and edited all of it and take full responsibility, as disclosed in the manuscript. I am the sole author. I have no competing interests to declare and the work received no funding.

Thank you for considering this revised submission.

Sincerely,

Leon Sandler